

Microscopy data analysis practical - smFISH

Philipp Savakis, Sander van Otterdijk, Evelina Tutucci

June 16, 2024

1 Overview

In this practical, you will explore and analyse microscopy image data from a single molecule fluorescence in situ hybridisation (smFISH) experiment on WT cells of *Saccharomyces cerevisiae* after the transition from 2% Raffinose or Glucose to 0.2% Galactose. This type of experiment allows the visualisation of individual mRNA molecules, and thus is a sensitive measurement of gene expression after these transitions.

To introduce you to the analysis of this type of data, you are handed fluorescence data from one location of a microscopy sample in the data/example folder. The data consists of three z-stacks, corresponding to the following channels:

- DAPI (bulk staining of nucleic acids - useful to find the location of the nucleus)
- CY3 (visualises a specific mRNA - which one it is, you need to infer from the data)
- CY5 (visualises another specific mRNA - which one it is, you need to infer from the data)

2 Setting up

2.1 required software and data

- Anaconda (install from website). You can install pycharm and/or spyder, but neither is required for this practical.
- Download napari_practical folder from canvas (contained in the 4_practical_smFISH2025.zip). This folder contains the code for the napari_plugin and several notebooks necessary in this practical.
- Download the smFISH_Selected Images_single_cell_course folder from canvas (contained in the 4_practical_smFISH2025.zip). The images contained in this folder are required to make representative overlays for your report later on.

2.1.1 navigation to correct folder

Under windows, hit the windows key and type anaconda. You should see Anaconda prompt

as a choice. Open anaconda prompt.

You should see a command line interface that looks something like:

```
(base) C:\Users\YourUserName> _
```

Use the command line to navigate to the unzipped data folder you downloaded from canvas. There, navigate to the napari_practical folder. Type

```
dir
```

to explore the directory's contents. You should find a file named requirements.txt
On mac, navigate to the napari_practical folder in the group folder you downloaded from canvas. Right-click on the folder and select open in new terminal window. You should see a command line interface that looks something like:

```
(base) User@user: ~/whereever/you/donwloaded/the/folder/to/napari_practical $ _
```

2.1.2 activation of environment

The (base) in your command line interface tells you that you are in the base environment - we have not activated our custom environment yet. Now, activate the quant environment by typing:

```
conda activate quant
```

On, windows, your command line should now show

```
(quant) C:\The\location\of\your\group\folder> _
```

On mac, you should see:

```
(quant) User@user: ~/whereever/you/donwloaded/the/folder/to/napari_practical $
```

We will now make use of pip for installation of most packages. Convince yourself that the folder you have navigated to contains a file named 'pyproject.toml'.

Now, to install the napari plugin for this course, type:

```
pip install -e .
```

Then, open napari, by typing:

```
napari
```

If all is well, you should be looking at this window:

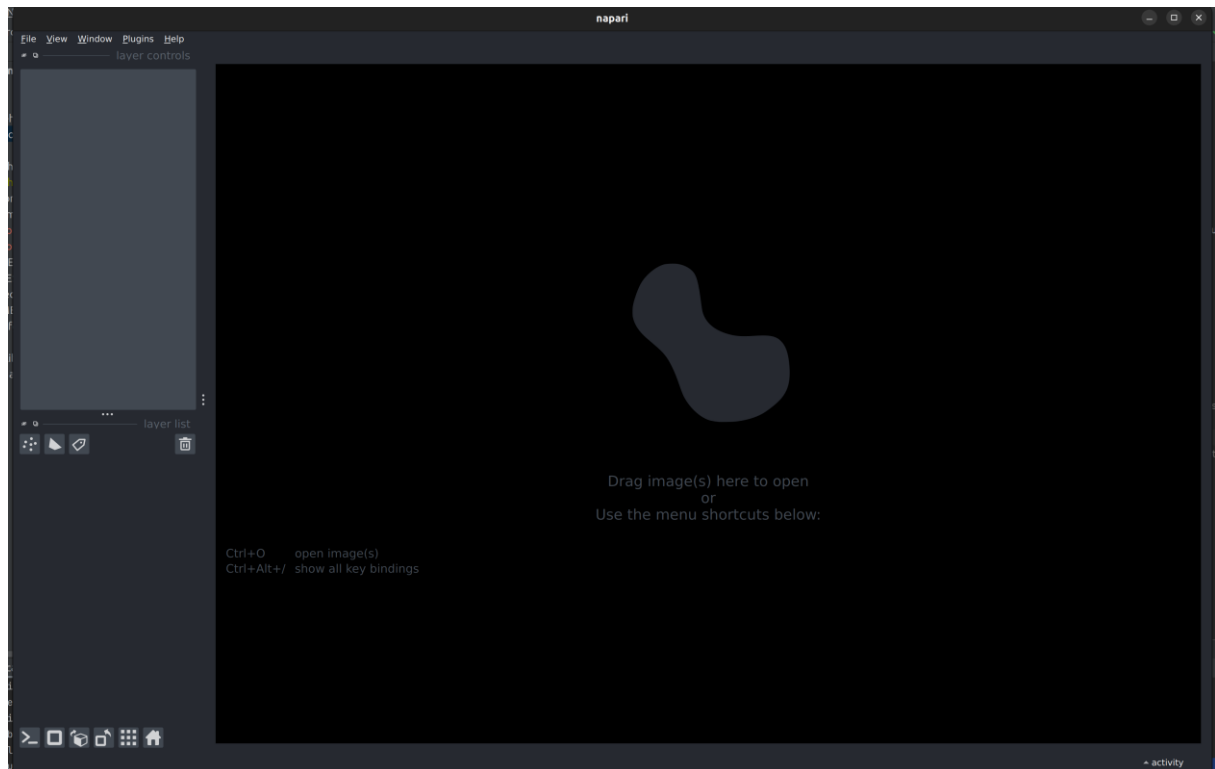


Figure 1: napari user interface

3 data exploration with napari

First, let's look at the data that was provided to you. Open the images by dragging them into the napari viewer. each .tif file in your /data/example/ folder has the following dimension: 41 x 736 x 736. This means that each image consists of 41 images, with height and width of 736 pixels each:

You can visualise different z-layers through the slider at the bottom of the image.

By default, napari makes no distinction between distances in the x,y,z direction.

Most microscopes, however, have lower resolution in z than they do in x/y. This means that two pixels in z direction are further apart than two pixels in x/y direction.

To tell napari that this is the case, we need to adjust layer scaling. We can do this by pressing the >_ button on the bottom left of the napari interface.

This should open a console window. In there, type:

```
viewer.layers[0].scale = [3, 1, 1]
viewer.layers[1].scale = [3, 1, 1]
viewer.layers[2].scale = [3, 1, 1]
```

Napari has powerful 3D visualisation functionality, which will allow us to explore the data in 3D. Press the square button next to the console button (or use the shortcut ctrl+y) in the napari interface. Now, the image should be transformed into a 3D image. Use your mouse to move around in the image. Use the menu to the left to manipulate colour representations and transparency/opactiy of the data. Can you create a 3D overlay, which shows the mRNA spots and the nuclei. Take a screenshot for your report.

When you are done exploring the data, restore the uniform scaling by typing:

```
viewer.layers[0].scale = [1, 1, 1]
viewer.layers[1].scale = [1, 1, 1]
viewer.layers[2].scale = [1, 1, 1]
```

into the napari console. Exit 3D visualisation mode through ctrl+y.

4 image segmentation with napari

In this practical, we want to say something about the number of transcripts in individual cells. To do this, we need to find individual cells first. To find them, we could label each individual cell by hand, but if we want to finish this practical today, we can use something slightly less labour-intensive:

- Find the Plugins menu item at the top of the napari user interface.
- Open the napari_practical/Random forest plugin.

Ensure that you are in 2D view mode. For computational reasons, we need to restrict cell segmentation to a single 736 by 736 image. We can generate this image from the z-stack, by generating a max projection. We will use the DAPI channel for segmentation.

So,

- select the DAPI layer
- adjust the z-layers you want to combine (z level start/finish)
- create the max projection with the max projection button.

This should create the following layers:

- Labels
- FISH_DAPI max proj

Now,

- Select Labels.
- Familiarise yourself with the interface on the left.
- use the drawing tools to label a part of the image.
- for the background use colour 1 (brown in the default colouring)
- for cell interior (nucleus and surroundings) use colour 2 (light blue in the default colouring)
- for the border between cell and background use colour 3 (purple in the default colouring)



Figure 2: interactive generation of training examples for random forest algorithm

Make sure that you use exactly these colors, or the underlying segmentation routine will do funny things.

Once you are happy with your artwork,

- select FISH_DAPI max proj
- hit the button classify selected layer

This creates a new layer named segmentation probabilities. This layer represents the random forest algorithm's belief that a given pixel in the image corresponds to a given class (background, cell interior, or border between cell and background). It might appear this layer is empty. **However, the output is stored only in the first three Z-stacks.** Given these probabilities, we need to now find regions in the image that belong to the same object. We will use the watershed algorithm to achieve this. The watershed algorithm needs seedpoints, which we can generate from the DAPI channel. The rationale is that high fluorescence in the DAPI channel corresponds to a nucleus, and nuclei are typically at the centroid of a cell.

Find the Plugins menu item at the top of the napari user interface.

- Open the napari_practical/segmentation plugin.
- select the FISH DAPI layer.
- adjust the intensity/distance parameters and press find nuclei
- remove superfluous layers (in case you had to use find nuclei more than once)
- check the nuclei layer you generated and add/remove points as needed
- select the segmentation probabilities layer
- click the watershed button
- this creates a segmentation layer.
- use the drawing tools to adjust cells as needed (Figure 3).
- save the segmentation image to disk (button on the right)

Saving the segmentation generates two files, one .tif file which holds the actual segmentation,

and a .pkl file, which holds information about all the unique cell labels. We need this information later to identify cells that are in the experiment but that have no mRNA counts.

5 Finding individual mRNAs

Normally, we would take the 3D data in all its glory and perform preprocessing and 3D gaussian fitting to capture point light sources (corresponding to single mRNAs) as precisely as possible. This is unfortunately beyond the capability of most laptops. Therefore, we have implemented a 2D spot finder for this practical.

- Find the Plugins menu item at the top of the napari user interface.
- Open the napari_practical/ spot finder plugin.
- select the FISH CY5 layer and create a max projection (use Random Forrest plugin)
- discard the Labels layer that was created here

Now, to make individual spots more prominent, we will use a trick, in which the

image is copied, and gaussian filters with different standard deviations are run over the individual copies. Then the difference between the images is computed (difference of gaussian, DoG).

You can adjust the gaussian filter through the sigma parameters, and then hit compute difference of gaussian. Play around with the values and inspect the resulting images.

Now, after having transformed the image with DoG, look at pixel intensities.

Select the difference of gaussian layer

- move your cursor over the image
- at the bottom left of the napari user interface, you should see numbers corresponding to the xy position, and to the intensity at a given point
- find, which values are typical for background pixels
- do the same for spot pixels
- find a threshold value inbetween two separate spots from background
- hit the threshold button
- adjust the newly created coords layer as needed, e.g.:

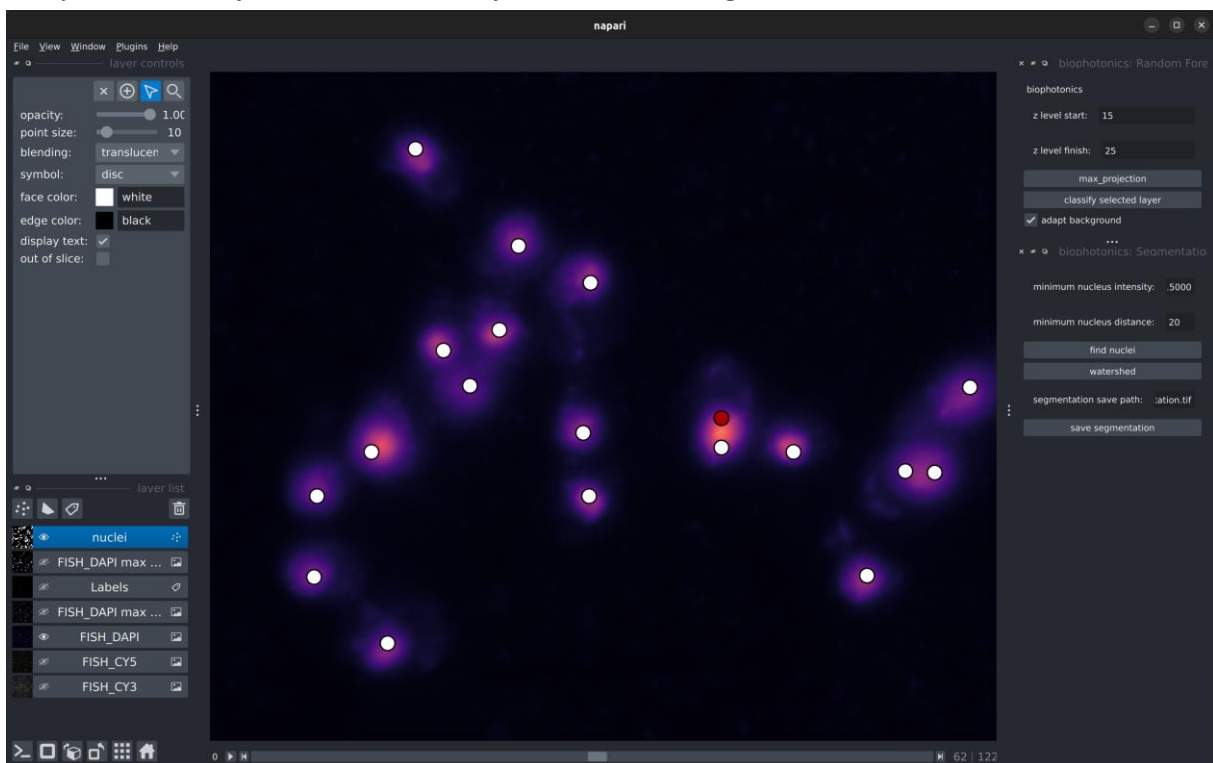


Figure 3: nuclei coordinates after nuclear detection. Sometimes two spots are identified.

Adjust these as needed.

now, ensure that all the right layers are selected: spot-intensity layer should be your CY-max projection, and spot position layer is the layer that you created during thresholding.

- save the spots to a pkl file using the save spot information button

- repeat this section for the other channel, FISH CY3.

6 data processing

You should now be the proud creator of two .pkl files. These files are in a storage format that allows data/object exchange between different python instances. Your file system likely will not know what to do with them.

To look at the content of your files,

- open a new instance of anaconda prompt (windows) or terminal (mac/linux)
- no need to activate the napari environment here - the base environment should have all we need
- navigate to the folder notebooks in the napari_practical folder you downloaded from canvas
- type jupyter notebook
- open 01_data_plotting.ipynb
- follow the steps laid out in the notebook.

In this notebook, you will plot the mRNA spot count data you have just extracted from the raw images in Napari.

7 data processing of provided data

Next, open 0.2_data_cleanup.ipynb. In this notebook we will provide you with RNA spot count data that we have extracted similarly already for you. This data can be found in two Excel sheets named :

- 1) FISH_results_GAL1mRNA_all_conditions.xlsx
- 2) FISH_results_DOA1mRNA_all_conditions.xlsx

The first sheet has expression data of the GAL1 mRNA. The second sheet has expression data of the DOA1 mRNA

Both sheets hold data for a total of four conditions:

- GLU - glucose only
- RAFF - raffinose only

- GLUGAL - 3 h after transition from glucose to galactose
- RAFFGAL - 30 min after transition from raffinose to galactose

We will first perform some data wrangling in this notebook. However, in the next (03 smFISH data visualisation.ipynb) notebook we will start plotting and analysing the provided FISH data as well.

7 The report

For the report, it is important you :

- Create representative overlay images of all FISH conditions for the report (see folder smFISH_Selected Images_single_cell_course.zip on canvas).
- Use the Napari plugin to extract the number of RNAs for one of the FISH transitions and compare it to the provided data in the two Excel sheets by plotting both. The images can be found in the folder data/smFISH_Selected Images_single cell course.
- Additionally, make sure to compare the FISH results to previously used methods' results in this course. For example, how does the FISH data compare to the flow cytometry data?